

Table 1: Transformation Rule Patterns

TR	Transformation Rule Pattern
CTR	$\Psi : C(s) \leftarrow R(r), URI_R(r, s), \delta(r)$ where: - Ψ is the name of the transformation rule; - R is the name of a relation or a view in the source schema $\mathbf{S}$; - r is a tuple variable bound to tuples of R; - s is the URI associated with the tuple bound to r; - $URI_R(r, s)$ associates the tuple bound to r to the URI s; - $\delta(r)$ is an optional selection over R.
OTR	$\Psi : P(s, o) \leftarrow R_D(d), B_D[d, s], R_G(g), B_G[g, o], \varphi(d, g)$ where: - Ψ is the name of the transformation rule; - Ψ is an object property in V with domain D and range G; - "$R_D(d), B_D[d, s]$" is the RHS of the CTR Ψ_D that matches class D with pivot relation R_D of Ψ_D; - "$R_G(g), B_G[g, o]$" is the RHS of the CTR Ψ_G that matches class G with pivot relation R_G of Ψ_G; - φ is a path from R_D to R_G; - $\varphi(d, g)$ holds iff d is connected to g following path φ.
Local DTR	$\Psi : P(s, v) \leftarrow R_D(r), B_D[r, s], nonNull(r.A_1), \dots, nonNull(r.A_k),$ $RDFLiteral(r.A_1, "A_1", "R_D", v_1), \dots, RDFLiteral(r.A_k, "A_k", "R_D", v_k),$ $T([v_1, \dots, v_k], v)$ where: - Ψ is the name of the transformation rule; - P is a datatype property of V with domain D; - "$R_D(r), B_D[r, s]$" is the RHS of the CTR Ψ_D that matches class D with pivot relation R_D of Ψ_D; - $A_1, \dots, A_k$ are attributes of R_D; - T is an optional function that transforms values of attributes $A_1, \dots, A_k$ of R_D to values of property P.
Path DTR	$\Psi : P(s, v) \leftarrow R_D(r), B_D[r, s], R_G(g), nonNull(g.A_1), \dots, nonNull(g.A_k),$ $RDFLiteral(g.A_1, "A_1", "R_G", v_1), \dots, RDFLiteral(g.A_k, "A_k", "R_G", v_k),$ $T([v_1, \dots, v_k], v), \varphi(r, g)$ where: - Ψ is the name of the transformation rule; - P is a datatype property of V with domain D; - "$R_D(r), B_D[r, s]$" is the RHS of the CTR Ψ_D that matches class D with pivot relation R_D of Ψ_D; - $A_1, \dots, A_k$ are attributes of R_G; - T is an optional function that transforms values of attributes $A_1, \dots, A_k$ of R_G to values of property P; - φ is a path from R_D to R_G; - $\varphi(r, g)$ holds iff r is connected to g following path φ.